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Reproduction and larval development of *Electrophorus electricus* on Marajó Island (Pará, Brazil)

Maria Ivaneide da Silva Assunção* and Horst O. Schwassmann**

Electrophorus electricus spawns during the late dry season, September to December, on Marajó Island. The breeding sites are residual water pockets inside root mats of groups of aninga plants (*Montrichardia arborescens*). Males construct foam nests and guard the growing larvae until mid-January when the first seasonal rains flood the breeding area, causing the ca. 10 cm long young eels to disperse. The growth of larvae was followed from the time of hatching in their nests by rearing them first in aquaria and subsequently in outdoor tanks. After 30 months, the largest specimens reached 70 cm. Electric eels reach sexual maturity when three years of age. Males outnumber females about three to one and are considerably larger than females. *E. electricus* is a partial spawner depositing up to three successive egg batches. Eggs and embryos from late batches serve as food for the first-born larvae. Not all eels with fully developed gonads inside the Goiapi drainage participated in the annual spawning activity suggesting that mating success depends in part on finding suitable breeding sites.

Na Ilha de Marajó, *Electrophorus electricus* se reproduz de setembro a dezembro, durante o período da seca. Os sítios de desova são poças residuais localizadas no interior dos aningaís (*Montrichardia arborescens*). Os machos constroem os ninhos com espumas e cuidam da prole até janeiro quando as primeiras chuvas sazonais inundam as áreas de reprodução, causando a dispersão dos filhotes. Nesse período os jovens medem mais ou menos 10 cm de comprimento. A taxa de crescimento das larvas foi registrada, iniciando-se nos ninhos e posteriormente em aquários e tanques de concreto. Após 30 meses o maior espécimen atingiu 70 cm de comprimento. A espécie alcança a maturidade sexual com três anos. Os machos são maiores e mais numerosos do que as fêmeas, cerca de 3 para 1. *E. electricus* possui desova parcial com até três sucessivas oviposições. Os ovos e os embriões resultantes das últimas desovas são utilizados como alimento pelas larvas nascidas nas primeiras desovas. No Rio Goiapi nem todos os poraquês conseguem se reproduzir no mesmo ano, apesar de apresentarem gônadas nos últimos estágios de maturação. A participação na atividade reprodutiva anual pode depender de encontrarem os locais adequados.

Introduction

Of the fresh-water electric fishes of South America (order Gymnotiformes), the electric eel *Electrophorus electricus* is the only species known to

produce powerful electric shocks, in addition to low-voltage orientation pulses. Its vernacular name in Brazil is 'poraquê' which originates from the Tupi language, meaning 'the one that stuns or debilitates'. The electric eel is easily recognized

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by its large size (up to two meters in length), its elongated shape and its coloration. While the larvae are uniformly black, the adults display an orange-red underside extending from the mandible over most of the abdomen.

Electrophorus has classically been considered a monotypic genus; a proposed second species from Peru, *E. multivalvulus* (Nakashima, 1941), is considered a synonym (Mago-Leccia, 1994). The earliest description in the European literature of this poraquê and its awesome shocking power seems to be by d'Abbeville (1614), a capuchin monk serving with French troops in the area that is now São Luiz, Maranhão.

Of the many scientific accounts about the electric eel, the voluminous posthumous treatise by Carl Sachs (1881), published by Emil du Bois-Reymond, lists all knowledge then available. Unfortunately, not much additional information concerning biology and natural behavior of this species has been added since. Aspects of reproduction and early development remained largely anecdotal and speculation. Embryonic and larval stages could not be procured in order to elucidate details of early electrogenic tissue formation. Accounts detailing the times of reproduction, spawning sites and behavior of *Electrophorus* did not exist. Attempts of artificial insemination by stripping of ripe specimens were unsuccessful (Sachs, 1881; Lissmann, in litt. 1965).

The work leading to the results reported here began in 1960 by the second author, who studied limnologic features and gymnotiform fishes of the Amazon estuary and Marajó Island at the Goeldi Museum at Belém (Egler & Schwassmann, 1962). First ecological data were obtained in 1961 when electric eels were collected for the Biophysics Institute in Rio de Janeiro where the biochemistry of electroplates has been studied since 1948 (Couceiro & Ackerman, 1948). The eel's electrosensory abilities were investigated at the Goeldi Museum in 1964 (Hagiwara et al., 1965; Szabo, 1966; Bullock, 1969). Joint field work by the present authors on Marajó Island began in 1983. Helped by local fishermen, we found the eel's breeding places in that area in 1988 and succeeded in observing reproductive behavior and rearing newly-hatched embryos through the larval period. Several notes on aspects of the eel's early development have already been published (Schwassmann & Assunção, 1989 a-c; Assunção & Schwassmann, 1992 a-b).

Material and methods

Marajó is this planet's largest estuarine island, covering an area of nearly 46,000 km² in the mouthbay of the Amazon. Most of the main river's discharge flows around its western and northern coasts, while its eastern borders are bathed by the outflowing waters of the Rio Tocantins in the Bay of Marajó, after receiving some contribution from the Amazon via the deep channels of Breves. This large island extends from the equator to almost 2° southern latitude, while stretching from 48°30' to 50°45' in western longitude. The western part of the island is of alluvial origin and is covered by seasonally flooded rainforest. The eastern half, however, is only partially of recent sedimentary origin and displays some high elevated terrain along stretches of the Marajó Bay coast. This eastern half is pasture land and undergoes an annual cycle of severe flooding by seasonal rains, alternating with a very dry period. The rains last from February to June, and are followed by a period of practically no rain between August and December when the land and most river courses become dry. When the eastern part of the island is covered by rain water of up to two meter depth, 'overland' travel is possible by boat only (Egler & Schwassmann, 1962; Sioli, 1966).

Audio-amplifiers with speakers and wire electrodes were used to localize eels. After a few occupied nests were detected, observation continued over several days during daylight hours. Samples of eggs and young were taken for size determination and for rearing in the laboratory. Nest sites were photographed and additional measurements were made by tape, ruler, and calipers of nests, spawn, and larvae. A Gossen foot-candle meter, a mercury-filled thermometer, and a Hanna conductivity meter provided measurements of physical characteristics of the spawning habitat and surrounding water bodies.

Five nests were utilized for observations during October-November 1988, seven nests were found and explored in November 1989, and two and three nests were investigated in 1990 and 1991 respectively. More than hundred newly hatched larvae from four nests were transferred with water and nest parts to small aquaria in the laboratory for an evaluation of growth rate.

Results

General observations about the electric eel population. We know very little of the eel's whereabouts during the rainy season when the entire Goiapi drainage is under water. Extensive surveys by visual means and with electronic amplifiers over many days during the rainy season of 1984 detected only a few eels in the central region of this area. This was also the case in the upper basin where there are some higher elevations and where we had been able to capture 115 large eels in mid-February of 1961. Fishermen only rarely succeed in catching an eel by cast net during the rainy season but from mid-August on, when the floods recede, eels appear to be abundant in certain deeper regions of the river course. We found them in dense aggregates at certain places on the banks and not at others, indicating social behavior. Inspecting more than a hundred adult eels during the middle and late part of the dry season, it appeared to us that, although all had fully matured gonads, not all eels participated in spawning activity every year. Many may not find suitable breeding places, a notion that is well supported by negative results of our own earlier attempts to initiate reproductive activity in captivity (unpublished protocol and results of 1983-84).

We have good evidence from frequent sampling that the sex ratio is skewed in favor of males. Males are much larger than females and have a more intensely colored red throat but small males may be easily mistaken for females. In late November of 1983 we measured, weighed, and dissected a sample of 40 adult eels. We found a ratio of 31 males to 9 females. Male average length was 152 cm (range 121-187 cm); average weight was 6.97 kg (range 4.3-12.3 kg). The nine females averaged 106 cm (range 95-115 cm) and their mean weight was 2.8 kg (range 2.0-3.7 kg). Gonads of all eels of both sexes were fully developed. Ovaries contained three overlapping size classes of immature, large, and yolky oocytes. Differences among males and females, other than size, are less striking; for example, a bulkier head and robust gape in males contrasts with a finer mouth in females. Females can attain sexual maturity at a smaller size than the lower limit of 95 cm indicated above. One eel raised by us was a mature female of 68.5 cm length at 3.5 years of age.

Time of reproduction. Reproductive activity consisting of nest construction and spawning was observed from September to November at our study site during four years of visiting the area in 1988-91. Fishermen reported that some eels were present at the nesting sites as early as August in some years when water levels were sufficiently low. From the size of larvae encountered in nests, we conclude that spawning does not begin before late September, reaching a peak probably in October. In late August of 1991, a quite unusual and relatively heavy rain occurred and raised the water level in the nesting area. During our November visit, we observed an advance of about two months in the onset of spawning activity of the *Hoplosternum thoracatum* population, also nesting in the same area. Normally, these catfish begin to spawn only in January at the onset of seasonal rains. The presence and activity of eels was considerably reduced at water levels higher than during the preceding three years.

Description of spawning sites. In January of 1988, breeding places of *E. electricus* were pointed out to us by local fishermen in residual backwaters of the Rio Goiapi about 12 km distant from the small town of Cachoeira do Arari (Fig. 1). Reproductive activity of *E. electricus* was encountered only inside a special inundated vegetation type that is characteristic for the interior of Marajó Island but is also known from other, mostly white-water floodplain regions. This so-called 'aningal' (plural aningais) consists of groups of densely spaced *Montrichardia arborescens* (family Araceae), a plant that can attain a height of up to 5 m and that is characterized by its large fan-shaped leaves (Figs. 2-3). On Marajó these plants are found growing along the margins of many river courses, resembling a ciliary, or gallery forest. They also grow on not yet solidified alluvial soil in permanently inundated depressions, known on Marajó as 'mondongos' (Miranda, 1908). At and below the water surface, these plants produce a mass of adventitious roots which intertwine and form a floating false floor of up to 80 cm thickness. At many places, this floor is interrupted by open-water pockets, pan-like basins that measure one to several meters in diameter.

Other fishes that find refuge in these aningais and also spawn here are *H. thoracatum* (tamoatá),

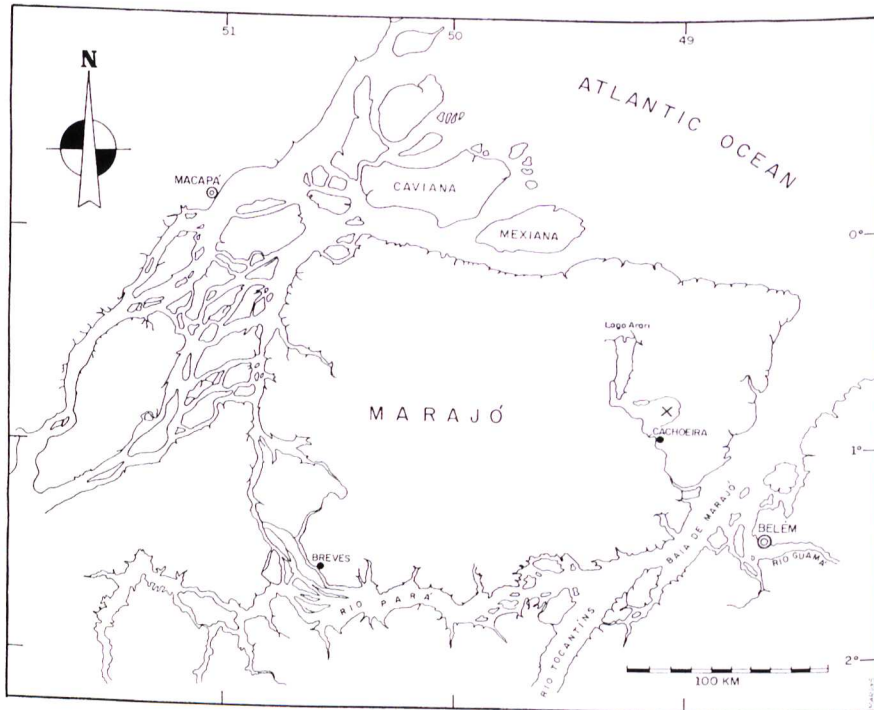


Fig. 1. Marajó Island with study site (X) near Cachoeira do Arari and adjacent rivers emphasized.

a callichthyid catfish, and *Brachyhypopomus pinicaudatus*, another gymnotiform knifefish. While the last species is very small and hardly noticed, the poraquê and tamoatá are noisily breaking the water's surface when they gulp air for respiration; hence these open-water pockets are called 'boiados' by fishermen. Eels seemed to prefer the smaller open-water areas. We measured eight of these boiados where eel nests were present and found a water depth of 50 to 90 cm, while the open water surface was one to three meters wide. Average temperature was 25.5°C ($\pm 1.0^{\circ}$); conductivity varied from 50 to $62 \mu\text{S}\cdot\text{cm}^{-1}$. Light levels inside the aningas were very low due to the dense spacing of aningas and their many large leaves. During midday hours and sunny skies, we measured an average of 33 foot-candles reflected from the water surface.

We could not ascertain any connections between adjacent boiados. Each nesting site was occupied by either only one pair of *E. electricus* or one male alone; females were more secretive and rarely seen.

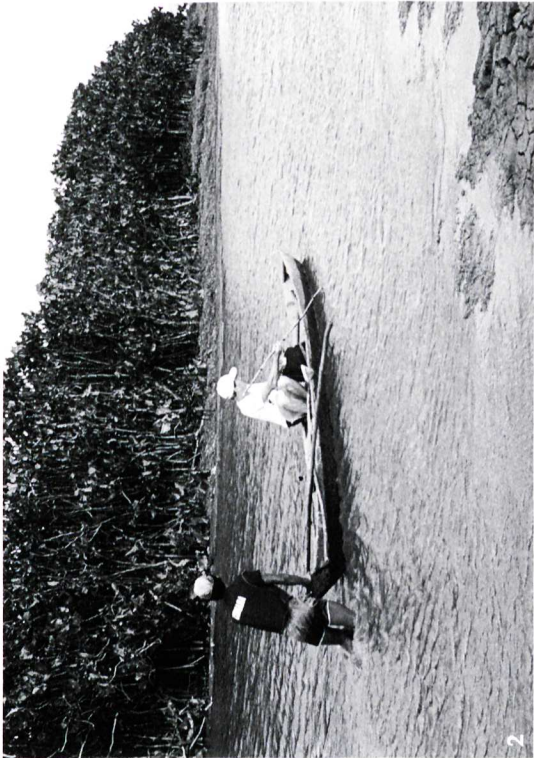
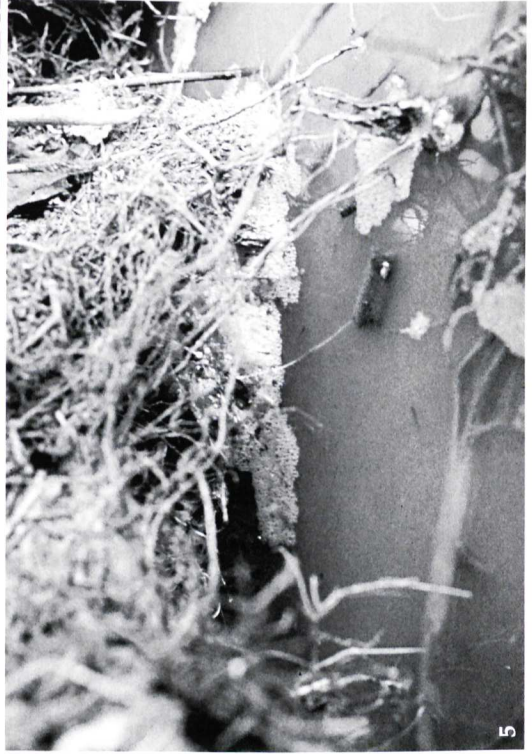
Nest construction. Males are building foam nests into which eggs are later deposited and fertilized. During nest-building behavior, males were seen to brake the water surface with their snout and, while slurping the surface layer, they moved the lower jaw rapidly for about one and a half minute at a frequency of 2.3-2.5 Hz (Fig. 4). Often, we could hear the noise made by the jaw movements. The male then swam to the darkest place of the boiado where fine adventitial roots were visible at the water surface. At this apparently preselected location, it expelled from four to eight tufts

Fig. 2. *Montrichardia arborescens* forming an 'aningal' near the nesting site in November, 1989.

Fig. 3. Our party entering the nesting area where three 10 cm long larvae were found on January 11, 1988.

Fig. 4. A male in its 'boiado' during 'foaming'. Surface ripples indicate high frequency of mandibular movements (lower right corner).

Fig. 5. Foam patches along the edge of the adventitial root mat during nest construction.



of foam (Fig. 5), after which it resumed normal respiration. The first foam patches expelled were always larger and more compact than the later ones. The earliest patches measured three to four cm in diameter, while the last ones were merely isolated air bubbles. The air bubbles making up the foam patches adhered to each other by a slimy substance; their size varied between one and five mm. Successive foam deposits occurred at irregular intervals. During the intervals we saw the male probe the entire nest area with its snout. More foam was added later in places where bubbles were less concentrated. The entire nest construction took from 12 to 17 hours. No plant substrate other than the aninga's adventitial root system was ever observed to be utilized for nest building (Fig. 6).

Spawning, hatching, and early development in the nests. Although we never witnessed the actual act of spawning and associated behavior, we have some indirect evidence that it occurs during daytime. Based on our findings of more than one size class of larvae present in some nests, and supported by the size classes of ovocytes in several dissected ovaries, we conclude spawning to take place in successive bouts during one breeding season. More than one batch of eggs can be deposited during one particular spawning bout. This is further supported by one of our long-term records from a nesting site where we followed the sequence of egg deposition and incubation. A five cm wide freshly deposited disk of eggs was noted at 09:00 hours on November 13, 1990. More eggs were found to have been added during our next visit at 16:00 hours this same day and disk size was now 17 cm. The egg patch remained undisturbed at this size until eight days later, November 21, when most embryos were observed to be hatching.

The disk-shaped egg mass was generally located in the approximate center of the foam nest. Those few eggs outside the foam and adhering to roots were found to have died after two to three

days. This was also the case with eggs which were transferred from the foam into small dishes containing water from the site. Sampling of eggs together with foam proved equally ineffective since the foam bubbles disintegrated within hours.

It was in this nest that we noticed the same pair to spawn repeatedly over a three months period. The first spawning must have taken place in early October, since the larvae taken out by fishermen in the second week of November were 40 to 50 mm long. The second egg batch was found on November 13, and these embryos hatched after eight days. While these larvae were growing, egg deposition must have continued until the end of December because we found new developing eggs among the older larvae. The December embryos did not survive but were eaten by the November group. We recorded this 'cannibalism' also inside another nest, and also during rearing experiments in the laboratory. In addition, spawn deposited by the female also contained many smaller undeveloped ovocytes and these, together with the later deposited developing embryos, appeared to be the preferred food of developing larvae.

Newly hatched embryos measured 8-11 mm, were transparent and rose-colored with a bulging yolk sac. Eye pigment and pectoral fin could at first not be discerned with the naked eye; and they also showed no reaction to bright light (Fig. 7). After three to four more days, the pink coloration became deeper and more solid, black eye pigment and pectorals were easily seen and oriented swimming movements were noted. Between five and eight days post-hatching, the larvae became dark colored, almost black, and they were observed to gulp air from the surface which made them buoyant, whereas before they always sank to the bottom when dislodged from foam. They were still found clumped together in the central area of the foam nest (Fig. 8). Nineteen days after hatching they measured 30-40 mm in length, were of black color and resembled small



Fig. 6. Nest with developing embryos seen from the open 'boiادouro' inside a recess in the root mat. The yellow color of foam is characteristic for older nests. The normally obscured recess has been enlarged for taking the picture.

Fig. 7. Recently hatched eel embryos.

Fig. 8. About three week old 30 mm long black larvae clumped together in the middle of the whitish foam nest.

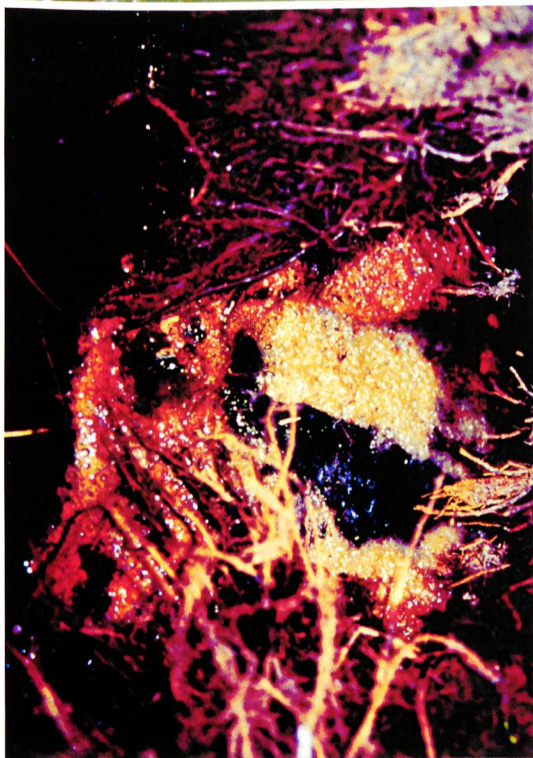
Fig. 9. After we had taken out the young from this nest, the resident male continued to be aggressive towards us, shocking repeatedly and almost climbing out of its nest hole.



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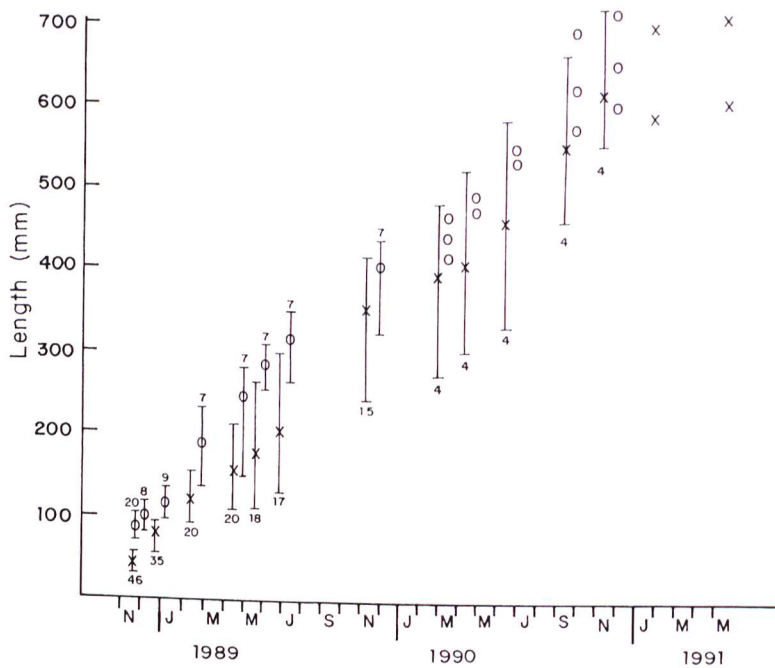


Fig. 10. Growth rate of young eels reared in captivity. Measured lengths of offspring from two nests (different symbols) are plotted with their means, range, and number of measured individuals. All points are shown when n was less than 4.

tadpoles by their wriggling swimming movements. They were now somewhat free of the foam nest, but always stayed together in clumps, usually in the darkest corners during rearing experiments in aquaria. Two months after hatching, the larvae had reached 70-90 mm. Already at a length of 55 mm, the larvae became less social and made many swimming ventures out of the group. Their swimming movements were no longer of the wriggle type and began to resemble more the kind of locomotion shown by older juveniles and adults, undulating the long anal fin while keeping the tail rigid. Stomach contents of 70 mm larvae consisted entirely of eggs, or developing embryos inside eggs; young of 80 mm had 80 % eggs and 20 % crustacean larvae in their stomachs, while 90 mm young had eaten less eggs and more crustacean larvae. Field observations were terminated at the end of January due to the onset of heavy seasonal rains which flooded the entire area.

Parental care. Not only do male electric eels build the foam nests, they also take care of nests and offspring. While females behave rather timidly

and flee or hide at the slightest disturbance, males are very aggressive during reproductive activity (Fig. 9). They are easily provoked into giving off trains of electrical discharges, so-called 'shocks', but always at a certain distance from the nest. We noted that every time that the nests were damaged, destroyed or young taken out by fishermen, the male constructed a new nest, modified it, or often relocated the former bubble nest. We also noticed one case of apparent brood-protecting behavior. When taking out several larvae by dipnet, the remainder of the young assembled over the head of the male which immediately swam with them to a deeper place under the root thicket, all the while the female was observed to quickly move away from the disturbance.

Growth rate. Since the larvae disperse and leave the nests when they measure 8-16 cm during January when rains inundate the aningais, the growth rate of individual fish cannot be followed in the field. We raised batches of very young eels, taken from different nests, in the laboratory, first in small aquaria, and later in larger outdoor concrete tanks. Fig. 10 shows the growth rate of two

of these groups over thirty months during which time they grew to almost 70 cm length. The fish seem to reach sexual maturity after three years of age. Females mature at a smaller size than males. Growth continues and males can reach two meters, or more. The total life span, in the absence of any known predators other than man, must be considerable. Male *E. electricus* have been kept in the ponds of the Goeldi Museum for more than 20 years.

Discussion

On Marajó Island the hydrological cycle limits growth for most fish species to about seven months of the year when most of the area is covered by a shallow layer of water. The land begins to dry up in August, and some of the larger fish species retreat to the lower reaches of the larger tidal rivers where water volume and reversing flow are maintained by tidal influx from the Bay of Marajó. But many species, including *E. electricus*, remain in the few residual water-filled pockets of smaller tributaries which by then no longer are connected to the larger rivers. These resident species are predominantly air breathers, thus being adapted to conditions of low oxygen levels in the extremely warm and overcrowded aquatic habitat. The majority of resident fish species seem to spawn at or after the onset of the rainy season in January-February (Schwassmann, 1992), while electric eels apparently are getting a head start over the mostly smaller species by breeding during the low-water season, September to November. Young electric eels will be more than 10 cm long when the rains have flooded the area and they will find ample food. The scarcity of food supply while inside the nest is made up for by the female eel in form of additional non-viable eggs, and by eggs and developing embryos that are deposited during subsequent egg-laying bouts.

The multiple, or partial spawning habit that has also been reported for several other gymnotoids (Kirschbaum, 1975, 1979, 1984, 1955; Schwassmann, 1976, 1992) has doubtlessly advantages in such a drastically fluctuating environment. For the electric eel this habit also serves as a means of providing food for the earlier batch of young. Construction of a foam nest and subsequent broodcare by the male helps in keeping the young close to their source of nourishment and also at

the water surface, permitting an early supply of air for respiration as well as for filling of the gas bladder. We do not know of any other case of parental care among the gymnotiform fishes. Mouthbrooding, as it is known for the aruanã and many cichlids, was not observed by us in the electric eel, although its possibility had been mentioned by Sachs (1881) implicating the buccal papillae, the real respiratory significance of which had escaped the earlier workers.

Conforming to its large size, *E. electricus* is of considerable longevity. No exact data seem available as to age; our mention of 20 years is certainly a conservative estimate. Reported maximum lengths of male electric eels range from 155 to 213 cm.

Although this report on reproduction and development of *E. electricus* should be followed up by more field studies to elucidate further details, it constitutes nevertheless the first account of breeding and early development of a gymnotiform species in its natural habitat. We lack information concerning the possible action of environmental factors in long-term advance timing of gonadal recrudescence and maturation, as it was shown for several smaller species in the laboratory (Kirschbaum, 1975, 1979, 1984, 1995).

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